DENISE AKASON

Reawakening the World's Most Famous Office Building:

Economics behind a Groundbreaking Energy Efficiency Retrofit

Anthony E. Malkin, president of Malkin Holdings, a New York City–based property asset management firm, looked out of a window from the 61st floor of the Empire State Building (ESB) at the Manhattan skyline on June 2, 2008. He was thinking about the presentation he had just heard from the team that had completed the first phase of project development on the plan that could change the world's view on whole-building energy retrofits, using ESB as the first test project.

Malkin's task was to reposition and rebuild ESB as a Class A trophy, and the team's plan, "The Empire State ReBuilding," would include creating one of the most energy-efficient and environmentally conscious office towers in the world. After listening to the presentation Malkin now had a clearer picture of the size, complexity, and risk of what he was trying to accomplish. He also had a glimpse of the potential rewards and dramatic changes to the prevailing view that showers and bike racks could make a building "green."

The report he had heard today was a good start, but Malkin knew much more had to be done before the promise could become reality.

Malkin Holdings

Malkin Holdings supervised more than 14 million square feet of office, retail, residential, and warehouse/distribution property in fifteen states. The portfolio contained 11 million square feet of trophy office property in the greater New York City area, including nine pre-war buildings (one of which was ESB) in midtown Manhattan, 1.9 million square feet of retail space, 1.4 million square feet of warehouse/distribution space, and 2,700 multifamily units.

ESB was acquired by Malkin's grandfather, Lawrence A. Wien; father, Peter L. Malkin; and Harry B. Helmsley in 1961. It is the world's most famous office building and has been reported by the American Institute of Architects as Americans' favorite example of architecture in the United States.[1] Its observatories attract more than 4 million visitors each year. The building stands 1,472

[1] Barbara J. Saffir, "AIA, Harris Interactive Poll: Empire State Building Tops the List of Beloved U.S. Buildings," *Architectural Record News*, February 7, 2007, http://archrecord.construction.com/news/daily/archives/070207aia.asp.

feet tall to the tip of its broadcast antennae and contains 2.85 million square feet of leasable office space.

Greening Commercial Buildings

In 2008 commercial, industrial, and residential buildings accounted for 40 percent of worldwide greenhouse gas emissions, with heating and cooling being the most energy-intensive activities, followed by lighting and other services. In New York City, buildings accounted for about two-thirds of the city's total carbon footprint, and older buildings were especially inefficient. Because 43 percent of all New York City office space (including the ESB) had been built before 1945, improving energy efficiency across the city posed a major challenge. As Malkin observed, "Constructing new green buildings won't move the needle in mitigating this problem. It's far more important to address the existing building stock."[2]

But some factors made energy efficiency a low priority for commercial buildings in the United States. First, energy represented a relatively small cost for most companies—one estimate put the average for a typical office-based company at 5 percent of total costs.[3] Second, the landlord-tenant relationship created conflicting interests regarding energy use. Additionally, the U.S. Green Building Council's LEED accreditation process[4] had oriented "green" around a scavenger hunt of finding points for flushless toilets, low-flow sinks, bike racks, showers, and "innovation points" for things like plant walls and water features—all items with questionable economic return for the investment made.

Additionally, many tenants were reluctant to invest in energy reduction because they would enjoy the benefits only through the end of their leases. On the other hand, landlords were hesitant to make energy-saving investments if a substantial portion of the benefits accrued to tenants. Malkin Properties had already instituted several green practices across its portfolio, including green pest management, green cleaning, energy-efficient maintenance vehicles, recycling of construction debris and tenant waste, use of recycled-content materials in floor coverings, tenant education, and improved air quality for tenants. It also measured and reported the energy efficiency of its buildings as a participant in the Energy Star program sponsored by the U.S. Environmental Protection Agency and Department of Energy.

ESB Project Objectives

Making ESB energy efficient was environmentally responsible, but now Malkin could see the promise of solid business reasons for undertaking such an audacious project. He wanted to prove the economic viability of whole-building energy efficiency retrofits. Reducing energy usage made sense in an era of rising costs and volatile markets, and if sustainable space could command higher

[2] Jonathan A. Schein, "Q&A: Tony Malkin, President of Wien & Malkin Properties," Expert Q&As, *Green Real Estate Daily*, June 22, 2008, http://www.metrogreenbusiness.com/news/qa.php/2008/06/22/p1245.

[3] Fiona Harvey, "Efforts Increase to Improve Sustainability," *Financial Times*, April 27, 2009, http://www.ft.com/intl/cms/7a011474-3028-11de-88e3-00144feabdc0.pdf.

[4] LEED (Leadership in Energy and Environmental Design) is a suite of rating systems for the design, construction, and operation of green buildings (for both existing and new construction).

rents and create a competitive edge in attracting companies that valued sustainability, it would result in higher occupancy and enhance a building's long-term value.

Malkin expected the project to enhance income in five ways:

1. Reduce existing capital improvement program costs
2. Lower the total utilities budget through more efficient use of energy and water
3. Reduce building maintenance and repair costs
4. Increase rent and occupancy by offering updated services, improving tenant perception, and appealing to tenant desire for green office space
5. Generate additional income from new tenant service offerings such as chilled water and emergency power

Economically, the retrofit could also be viewed as a hedge against the possibility of future regulation that could mandate reductions in greenhouse gases.

If successful, Malkin and his initial partner, the Clinton Climate Initiative, wanted to use the ESB project to develop a model for whole-building retrofits that could demonstrate that energy efficiency with a defined economic return could be replicated in other buildings, especially pre-war buildings. "The goal with the Empire State Building has been to define intelligent choices, which will either save money, spend the same money more efficiently, or spend additional sums for which there is reasonable payback through savings," he said. "Addressing these investments correctly will create a competitive advantage for ownership through lower costs and a better work environment for tenants."[5]

Project Team

Malkin's first step was to assemble a team of consultants in the fields of climate change, real estate sustainability, environmental design, and energy services. The idea was first brought to him by Jamie Russell of the Clinton Climate Initiative. In addition to ESB Operations, the team consisted of representatives of the Clinton Climate Initiative, Johnson Controls, Jones Lang LaSalle, and the Rocky Mountain Institute. **Exhibit 1** briefly describes each organization and its role in the project.

Shortly after it was formed in April 2008, the project team agreed on a clear charter: "The retrofit of the Empire State Building into a Class A pre-war trophy building will transform the global real estate industry by transparently demonstrating how to create a competitive advantage for building owners and tenants through profitably greening existing buildings."

The team established a four-phase project development process that commenced in April 2008 and ended in November 2008 (see **Exhibit 2**). Project development included understanding current

[5] Empire State Building Sustainability Program, http://esbny.com/documents/sustainability/ESBPlacematFINAL2.pdf (accessed January 12, 2012).

performance, analyzing opportunities to improve performance, and determining which projects to implement.

Inventory and Programming Phase

The key deliverable of the first phase—inventory and programming—was a review of how project goals and standards could be integrated with $93 million in 2008 capital projects that were planned or underway. ESB had already initiated planning for a $550 million program of restoration and upgrades to the observatories, art deco masterpiece lobby, hallways, restrooms, and other common areas. The intent of the integration was to identify opportunities to integrate energy efficiency into the planned capital projects; if successful, this whole-building approach could make the building greener with modest increases in the current capital budget.

During this phase the team also reviewed all ESB mechanical systems and equipment and calculated tenant energy usage, an effort that was part of developing a baseline energy benchmark and system for measuring energy efficiency. (See **Exhibit 3** for a summary of ESB energy consumption at the outset of the project.) In addition, the team scoped out how the final product would perform against a LEED assessment to determine which criteria ESB was already meeting and which others were feasible during the course of the project.

Design Development Phase

During the second phase—design development—the team first planned to focus on measuring how much electricity and steam each tenant used and for what purpose, whether broadcast, radiator heating, lighting, or cooling. The team would also develop system-level energy usage statistics along with a theoretical minimum level of energy use to help evaluate the potential for future reduction.

The team would begin to engage tenants by implementing a tenant energy program during this phase. Each tenant's energy usage would be submetered and displayed on the ESB webpage along with benchmarking information (see **Exhibit 4**). Building staff would train program contacts for each tenant and recommend best practices, and each tenant would report on progress in reducing energy consumption. This approach would ensure that tenants understood how the retrofit would benefit them economically.

In addition, in order to attract new tenants the team planned to design sustainable pre-built spaces ready for move-in. These spaces would use limited wall enclosures to admit natural light, interior finishes produced in a sustainable manner, task lighting that enabled higher-efficiency overhead lighting, and more efficient mechanical systems. Designing the spaces would enable ESB to articulate the costs and benefits of the green design and demonstrate design principles for all tenants. The team also anticipated providing tenants with design guidelines based on green standards along with tools to validate their economic benefits.

Design Documentation Phase

The third phase of the analysis would produce a final report documenting tenant energy usage, which would establish the building's energy consumption baseline. The team also planned to develop and refine a model (eQUEST) to compare projected improvements with this baseline in order to calculate energy savings—not only on a standalone basis but also in various combinations with other energy-saving measures. The team would use this information to determine how each new measure would affect project costs and building performance.

The team planned to use the LEED rating systems both for existing buildings (EB) and for commercial interiors (CI) by creating LEED EB/CI feasibility reports to assess the levels of certification that could reasonably be achieved by the base building itself, as well as for tenants who desired additional certification for their own space. Being able to tell tenants during the request for proposal phase that they could reasonably achieve LEED CI Silver at a quantified premium, for example, would help them decide if formal certification was right for them.

Another essential part of this phase was conducting in-depth lease reviews and tenant surveys. The surveys would help tenants stay engaged in the sustainability process and ensure their needs were considered during the project. Understanding the timing of lease renewals and tenant attitudes regarding sustainability would help Malkin's management team make pricing decisions when renewing or starting leases.

Final Documentation

The main deliverable of the final phase of project development was the integrated sustainability master plan report, which included data from all standards and measurement tools, including the eQUEST energy-modeling tool. The report would describe in detail the measures planned—and benefits expected—in a format that could be easily understood and replicated by owners of other pre-World War II buildings. **Exhibit 5** shows the list of projects considered.

The team made several key assumptions in order to model the costs and benefits for the base case of the project over a fifteen-year time horizon:

Annual fuel escalation	=	1%
Annual construction escalation	=	1%
Annual inflation	=	2%
Real discount rate	=	8%
Green rent premium	=	1%

For each project (or package of projects) the team would calculate financial benefits measured by net present value (NPV) and environmental benefits measured by carbon dioxide (CO_2) reductions. During this phase the team would recommend how to balance the tradeoff between the two—at one extreme the team could maximize NPV and at the other it could maximize CO_2 reductions. Other options were maximizing CO_2 savings for a zero NPV, or balancing NPV and CO_2 reduction. **Exhibit 6** shows the tradeoff between these two objectives.

Financing

Malkin planned to finance the ESB project from cash flow, but explained his interest in financing options, including the energy service company model described in Exhibit 1: "We will be working to establish a financing format to provide the ability to otherwise indebted properties to participate in this sort of project, though the work on this project is not financing contingent and is going forward out of already available cash."[6]

Conclusion

Malkin wrote down some questions to discuss with the team leaders as they kicked off the second phase of project development.

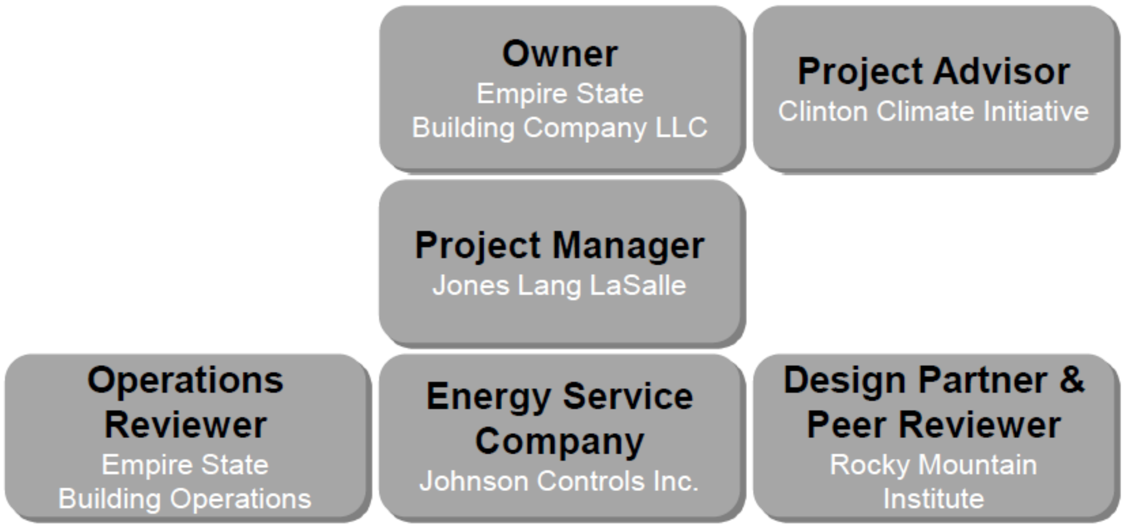
First, how should they decide which energy reduction projects to pursue? A closely related question was how Malkin should determine the right tradeoff between financial return and CO_2 reductions. Malkin also wanted the team leaders' input about the potential risks of a project of this magnitude and complexity, and what they recommended to minimize them.

Malkin turned his gaze again to the world-famous skyline and smiled. "We'd better do this in secret," he thought to himself. "We might prove that this can't work . . . but if we prove it can, we will change the world." This was going to be fun.

[6] "Jones Lang LaSalle Manages Landmark Empire State Building Sustainability Program to Reduce Energy and Carbon by 38% and Serve as Industry Model," press release, April 6, 2009, http://www.us.am.joneslanglasalle.com/Pages/NewsItem.aspx?ItemID=9178.

Exhibit 1: Empire State Building Integrated Energy Retrofit (IER) Project Team

TEAM ORGANIZATION CHART



DESCRIPTION OF PARTNERS

Empire State Building Operations was the site champion and had the task of ensuring that ongoing operations and tenant comfort and safety were not disrupted.

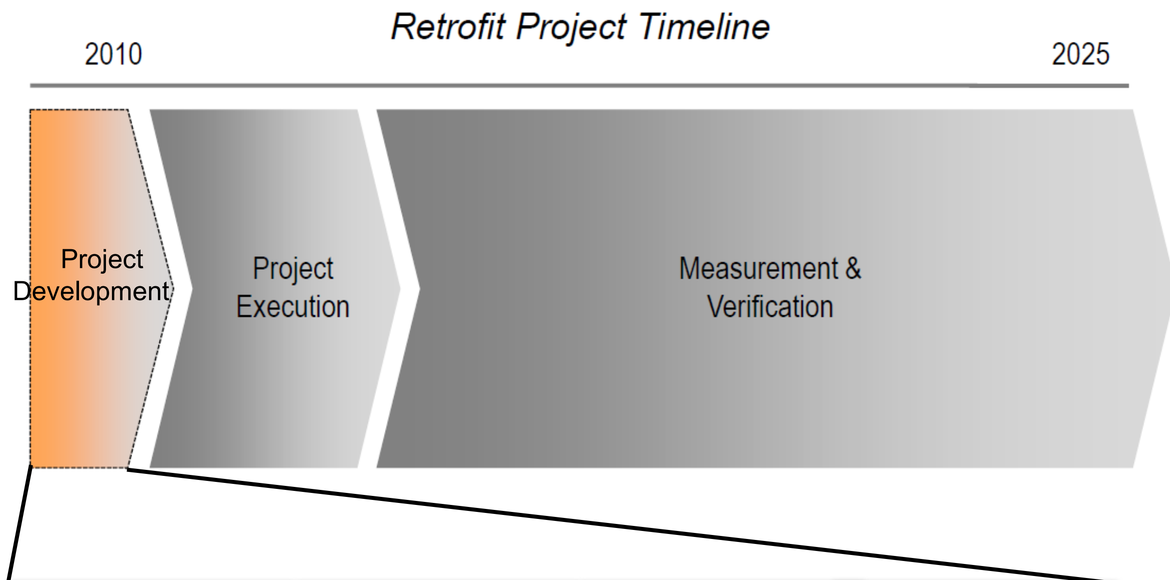
The Clinton Climate Initiative was founded in August 2006 by the William J. Clinton Foundation to create solutions to the core issues driving climate change. The Clinton Climate Initiative worked with ESB owners to reduce greenhouse gas emissions and develop and validate the owner's vision.

Johnson Controls, Inc., a global Fortune 100 company, served as the energy service company (ESCO) for the project and performed engineering, procurement, and construction work under a performance-contracting framework that guaranteed savings and provided long-term energy management. A building owner could finance sustainable projects with little or no money up front based on the cash flow from reduced energy costs as guaranteed by an ESCO partner.

Jones Lang LaSalle, a global real estate services firm, acted as the program manager for the ESB project. Members of its energy and sustainability services group led the team through the collaborative process involved in an IER project. The program manager was responsible for ensuring team collaboration, stakeholder communication, and timely execution, as well as driving performance measurement and documenting the model for industry-wide use.

The Rocky Mountain Institute, a nonprofit leader in energy-efficient solutions, provided vital expertise and conducted peer reviews on technical and design elements of the energy work in the building.

Exhibit 2: Project Timeline



	Phase I: Inventory & Programming	***Phase II: Design Development***	***Phase III: Design Documentation***	***Phase IV: Final Documentation***
Activities	• April 14 kick-off meeting • May 7–14 team workshops • June 2 presentation to ownership	• June 18 theoretical minimum workshop • July 2 workshop • July 15 presentation to ownership	• July 30 tenant focus workshop • August 13 eQUEST workshop • August 27 presentation to ownership	• Sept. 10 workshop • Sept 29 presentation to ownership • Oct. 6–8 finance workshop • Nov. 10 presentation to ownership
Outputs	• Baseline capital projects report	• Baseline energy benchmark report	• Tenant initiatives report (pre-builts, design guidelines, energy management) • Tuned eQUEST mode	• Model (eQUEST, financial, greenhouse gas) outputs • Integrated sustainability master plan report (including energy master plan)

Source: “Empire State Building Case Study: Cost-Effective Greenhouse Gas Reductions via Whole-Building Retrofits,” http://www.esbnyc.com/documents/sustainability/ESBOverviewDeck.pdf (accessed January 12, 2012).

Exhibit 3: Empire State Building before IER Project

102 stories and **2.8 million** square feet
4.0 million visitors per year
$11 million in annual energy costs
Peak **electric** demand of **9.5 MW** down from 11.6 (3.8 W/sf incl. HVAC)
88 kBtu per sf per yr for the office building
CO_2 emissions of **25,000 tons** per year (22 lbs/sqft)

Source: "Empire State Building Case Study: Cost-Effective Greenhouse Gas Reductions via Whole-Building Retrofits," http://www.esbnyc.com/documents/sustainability/ESBOverviewDeck.pdf (accessed January 12, 2012).

Exhibit 4: Sample Tenant Energy Use Webpage

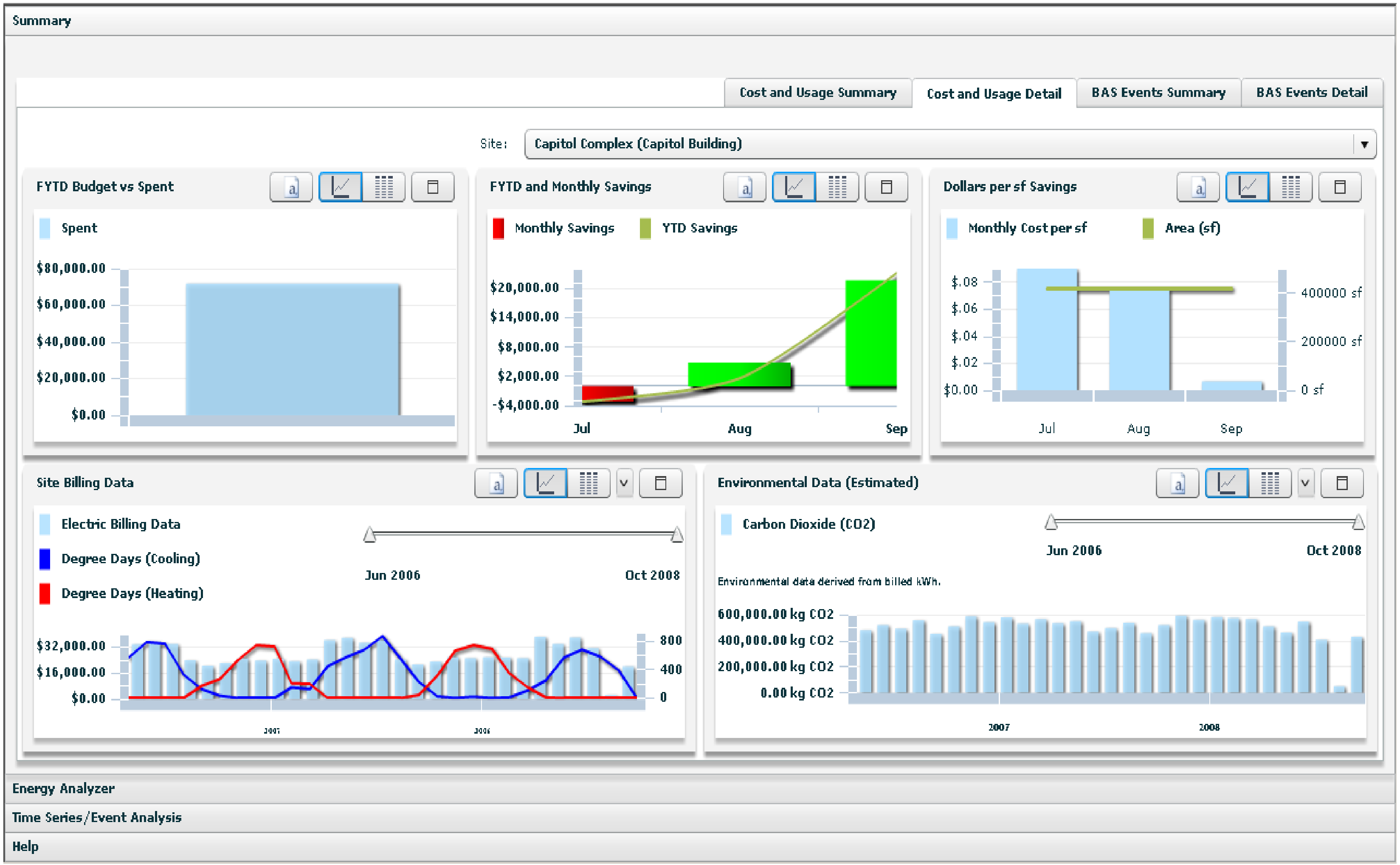


Exhibit 5: Summary of All Projects Investigated by ESSB Team

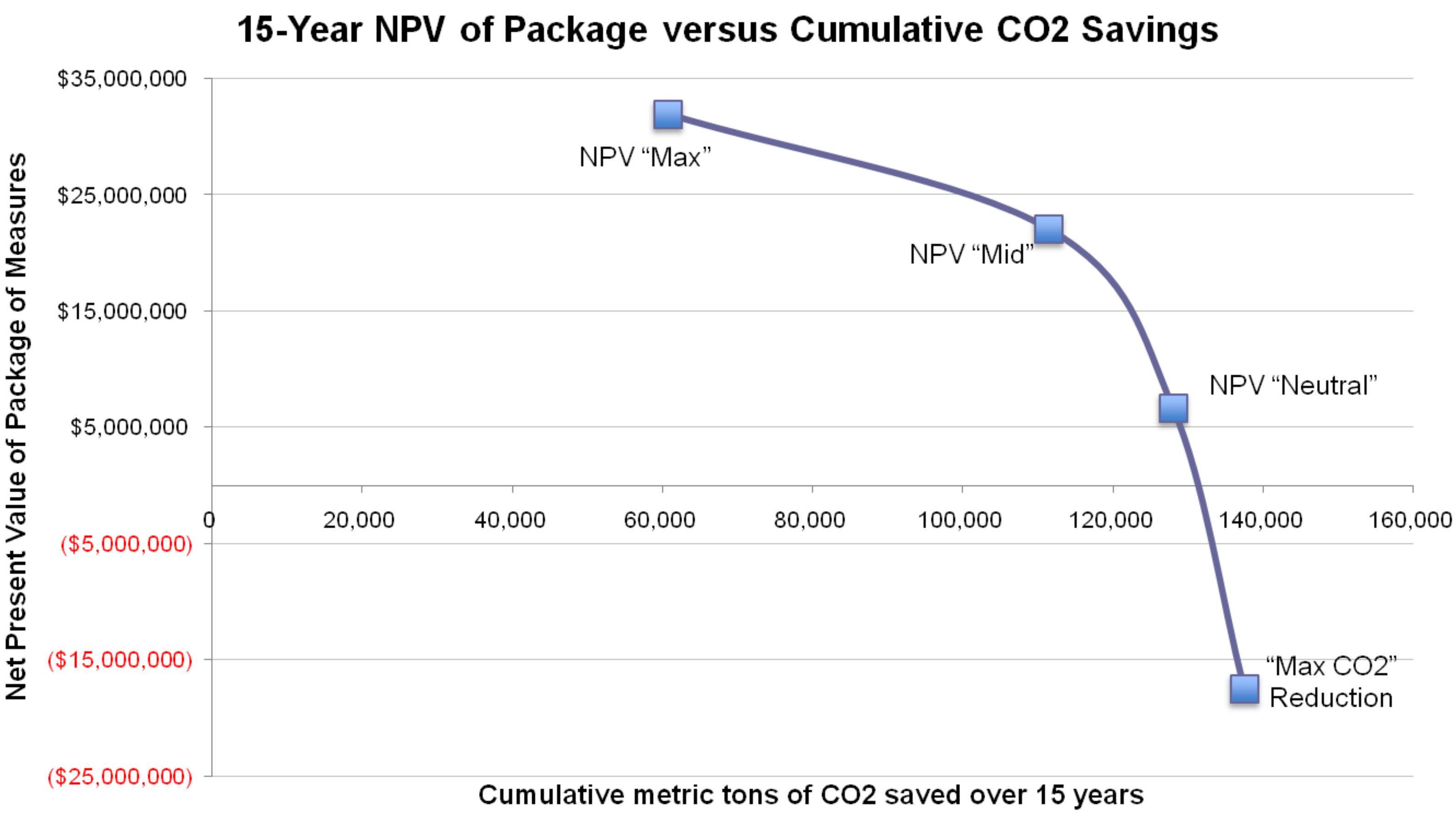
Project		Incremental or Add to 2008 Program	ESSB Budget	2008 ESB Capital Budget	Incremental Cost	Annual Energy Savings (Once Fully Implemented)
1	Building Windows	Incremental to planned ESB window refurbishment project	($4,458,768)	($455,000)	($4,003,768)	$368,343 (JCI)
	T-Operative	Incremental to planned ESB window refurbishment project	Included Above	Included Above	Included Above	$41,537 (Tenants)
2	102nd Floor Windows	Add	($56,430)	$0	($56,430)	$2,629
3	Radiative Barrier	Add	($2,684,053)	$0	($2,684,053)	$190,212 (JCI)
4	Green Roof/Cool Roof	Add	($570,000)	$0	($570,000)	$4,221
5	Balance of DDC	Incremental to ongoing ESB controls upgrade project	($7,603,686)	($2,000,000)	($5,603,686)	$740,905 (JCI)
6	Tenant DCV	Add	Included Above	Included Above	Included Above	$117,400 (JCI)
7	Additional DDC Controls	Add	($2,000,000)	$0	($2,000,000)	$30,682
8	LED Tower Lighting	Add	($7,000,000)	$0	($7,000,000)	$33,575
9	Retrofit Chiller Plant	Incremental to proposed ESB chiller replacement project	($5,111,140)	($12,400,000)	$7,288,860	$675,714 (JCI)
	New Capacity and Riser	Incremental to proposed ESB chiller replacement project	Included Above	($10,000,000)	$10,000,000	Included Above
10	Corridor Lighting	Incremental to planned/ongoing ESB corridor upgrade project	($3,416,685)	($1,594,453)	($1,822,232)	$19,534
11	Restroom Lighting	Incremental to planned/ongoing ESB restroom upgrade project	($600,000)	($280,000)	($320,000)	$3,939

. . . . . .

Exhibit 5 *(continued)*

Project		Incremental or Add to 2008 Program	ESSB Budget	2008 ESB Capital Budget	Incremental Cost	Annual Energy Savings (Once Fully Implemented)
12	VAV AHUs	Incremental to ongoing ESB replacement of constant volume AHUs	($47,200,000)	($44,850,000)	($2,350,000)	$702,507 (ESB)
13	Tenant Daylight/Lighting/ Plugs	Incremental to what tenants typically install	($24,532,606)	($16,105,981)	($8,426,625)	$940,862 (Tenants)
14	Tenant Wall Insulation	Add	($7,740,000)	$0	($7,740,000)	$123,503
15	JCI Tenant Energy Management	Add	($365,734)	$0	($365,734)	$25,000 (JCI)
	ESB Tenant Energy Management	Add	Included Above	Included Above	Included Above	$361,709 (ESB)
16	Power Generation	Add	($14,969,992)	($7,871,040)	($7,098,952)	$320,000 (JCI)

Exhibit 6: NPV vs. CO_2 Savings



Source: “Empire State Building Case Study: Cost-Effective Greenhouse Gas Reductions via Whole-Building Retrofits,” http://www.esbnyc.com/documents/sustainability/ESBOverviewDeck.pdf (accessed January 12, 2012).